RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)RESOLUTION MEPC.397(83)
(adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES
(NOx TECHNICAL CODE 2008)
(Use of multiple engine operational profiles for a marine diesel engine,
including clarifying engine test cycles)
THE MARINE ENVIRONMENT PROTECTION COMMITTEE
RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee conferred upon it by international conventions for the prevention and control of marine pollution from ships,
RECALLING ALSO article 16 of the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocols of 1978 and 1997 relating thereto (MARPOL), which specifies the amendment procedure and confers upon the appropriate body of the Organization the function of considering amendments thereto for adoption by the Parties,

which specifies the amendment procedure and confers upon the appropriate body of the Organization the function of considering amendments thereto for adoption by the Parties,
RECALLING FURTHER regulation 13 of MARPOL Annex VI, which makes the Technical Code on Control of Emission of Nitrogen Oxides from Marine Diesel Engines (hereafter "NO x Technical Code 2008") mandatory under that Annex,
HAVING CONSIDERED, at its eighty -third session, draft amendments to the NOx Technical Code 2008 concerning the use of multiple engine operational profiles for a marine diesel engine, including clarifying engine test cycles, as appropriate, approved at its eighty-second session and duly circulated in accordance with article 16(2)(a) of MARPOL,
1 ADOPTS, in accordance with article 16(2)(d) of MARPOL, amendments to the NO x Technical Code 2008, the text of which is set out in the annex to the present resolution;
2 DETERMINES, in accordance with articles 16(2)(f)(ii) and (iii) of MARPOL, that the amendments shall be deemed to have been accepted on 1 September 2026 unless prior to that date not less than one third of the Parties or Parties the combined merchant fleets of which

amendments shall be deemed to have been accepted on 1 September 2026 unless prior to that date not less than one third of the Parties or Parties the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet have communicated to the Organization their objection to the amendments;

3 INVITES the Parties to note that, in accordance with article 16(2)(g)(ii) of MARPOL, the said amendments shall enter into force on 1 March 2027 upon their acceptance in accordance with paragraph 2 above;

4 ALSO INVITES the Parties to note that the said amendments shall enter into effect as follows:

(a) for a new individual engine or a parent engine of an engine family or engine group that has not been previously certified, the said amendments apply no later than 1 January 2028, based on the issue date of the EIAPP Certificate for the individual engine or parent engine;

(b) in the case of a new member engine to an engine family or engine group for

(b) in the case of a new member engine to an engine family or engine group for which the parent engine was certified before 1 January 2028, prior to the certification of that member engine it would need to be shown that the engine family or engine group complie d with the said amendments no later than 1 January 2030 based on the issue date of the EIAPP Certificate for that member engine;
(c) the said amendments do not apply to a marine diesel engine which already has an EIAPP Certificate except:
(i) in the case of an engine that is subject to substantial modification on or after 1 January 2028, the said amendments would apply as specified in the definitions of "substantial modification" set out in amended paragraph 1.3.2 of the NOx Technical Code 2008 based on the issue date of the EIAPP Certificate for that engine; and
(ii) in the case of an identical replacement engine installed on or after 1 January 2028, the version of the NOx Technical Code 2008 at the time of issuance of the EIAPP Certificate to the original engine

(ii) in the case of an identical replacement engine installed on or after 1 January 2028, the version of the NOx Technical Code 2008 at the time of issuance of the EIAPP Certificate to the original engine applies, unless the replaced engine is already equipped with multiple engine operational profiles, in which case the provisions of the new chapter 8 of the NOx Technical Code 2008 apply;

5 REQUESTS the Secretary-General, for the purposes of article 16(2)(e) of MARPOL, to transmit certified copies of the present resolution and the text of the amendments contained in the annex to all Parties to MARPOL;

6 ALSO REQUESTS the Secretary-General to transmit copies of the present resolution and its annex to Members of the Organization which are not Parties to MARPOL.

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ANNEX

AMENDMENTS TO THE NOx TECHNICAL CODE 2008

(Use of multiple engine operational profiles for a marine diesel engine, including clarifying engine test cycles)

Chapter 1 General

1.3 Definitions

ANNEX

AMENDMENTS TO THE NOx TECHNICAL CODE 2008

(Use of multiple engine operational profiles for a marine diesel engine, including clarifying engine test cycles)

Chapter 1 General

1.3 Definitions

1 Paragraph 1.3.2 is replaced by the following:

"1.3.2 Substantial modification of a marine diesel engine means:

.1 For engines installed on ships constructed on or after 1 January 2000, substantial modification means any modification to an engine that could potentially cause the engine to exceed the applicable emission limit set out in regulation 13. Routine replacement of engine components by parts specified in the technical file that do not alter emission char acteristics shall not be considered a "substantial modification" regardless of whether one part or many parts are replaced. For the recertification of such an engine following a substantial modification, the version of this Code that was used for the original certification shall apply except if the engine was or is now equipped with an auxiliary control device or has multiple engine operational profiles. Where an auxiliary control device is fitted, the requirements of 2.5 and 3.3 of this Code shall

engine was or is now equipped with an auxiliary control device or has multiple engine operational profiles. Where an auxiliary control device is fitted, the requirements of 2.5 and 3.3 of this Code shall apply. Where there are multiple engine operating profiles, the requirements of chapter 8 of this Code shall apply.

.2 For engines installed on ships constructed before 1 January 2000, substantial modification means any modification made to an engine that increases its existing emission characteristics established by the simplified measurement method as described in 6.3 in excess of the allowances set out in 6.3.11. These changes include, but are not limited to, changes in its operations or in its technical parameters (e.g. changing camshafts, fuel injection systems, air systems, combustion chamber configuration, or timi ng calibration of the engine). The installation of a certified approved method pursuant to regulation 13.7.1.1 or certification pursuant to regulation 13.7.1.2 is not considered to be a substantial modification for the purpose of the application of regulation 13.2 of the annex. For recertification of

regulation 13.7.1.1 or certification pursuant to regulation 13.7.1.2 is not considered to be a substantial modification for the purpose of the application of regulation 13.2 of the annex. For recertification of such an engine following a substantial modification, 2.5, 3.3 and, where that engine has multiple engine operating profiles, chapter 8 of this Code shall apply."

2 New paragraphs 1.3.21 to 1.3.37 are added as follows:

"1.3.21 Engine operational profile means a particular set of NOx influencing settings applied in the base emission control strategy which influences the NO x emission performance. Those settings may relate to, but are not limited to, fuel injection, inlet and exhaust valve operation, charge air management, exhaust bypass/wastegate or exhaust after-treatment controls and auxiliary control devices.

1.3.22 Multiple engine operational profiles means that more than one engine operational profile is available for selection on a marine diesel engine.

OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 4

1.3.22 Multiple engine operational profiles means that more than one engine operational profile is available for selection on a marine diesel engine.

1.3.23 Auxiliary control device means a system, function or control strategy installed on a marine diesel engine that is used to protect the engine and/or its ancillary equipment against operating conditions that could result in damage or failure, or that is used to facilitate the starting of the engine. An auxiliary control device may also be a strategy or measure that has been satisfactorily demonstrated not to be a defeat device. An auxiliary control device includes any element of design that includes sensors, or other arrangements which, by an action of the control system, can activate, modulate, delay or deactivate the operation of any part of the base emission control system. Any device or strategy the activation of which causes a nonprogressive change in emissions is also an auxiliary control device. An auxiliary control device not declared at the time of the first certification of a marine diesel engine

nonprogressive change in emissions is also an auxiliary control device. An auxiliary control device not declared at the time of the first certification of a marine diesel engine shall be considered a defeat device.

1.3.24 Defeat device means a device that measures, senses or responds to operating variables (e.g. engine speed, temperature, intake pressure or any other parameter) for the purpose of activating, modulating, delaying or deactivating the operation of any component or the function of the emission control system such that the effectiveness of the emission control system is reduced under conditions encountered during normal operation, unless the use of such a device is substantially included in the applied emi ssion certification test procedures. An auxiliary control device accepted as part of the Administration's review of the NOx certification pack is not a defeat device.

1.3.25 Base emission control strategy means the emission control strategy active at any time an auxiliary control device is not active. It consists of any parameter, element of design, or operating control that is designed to modulate as a function of

at any time an auxiliary control device is not active. It consists of any parameter, element of design, or operating control that is designed to modulate as a function of engine load and/or speed in a manner that affects the emission performance of the engine. The modulation of parameters is to be progressive and shall not result in a disproportionate change in emissions.

1.3.26 Rational emission control strategy means the base emission control strategy applied to a marine diesel engine which ensures that the emission values at the individual mode points as used to give the weighted specific emission value are representative of the emission values during normal operation of the engine.

1.3.27 Irrational emission control strategy means any strategy or measure that, when a marine diesel engine is operated under normal conditions of use, reduces the effectiveness of an emission control system to a level below that expected from the applicable emission test procedures.

1.3.28 Not-to-exceed emission limit value means the maximum permitted NO x emission value at a given operating condition as determined in accordance with 3.3

applicable emission test procedures.
1.3.28 Not-to-exceed emission limit value means the maximum permitted NO x emission value at a given operating condition as determined in accordance with 3.3 of this Code within the not-to-exceed zone of the engine.
1.3.29 Not-to-exceed zone means the power or torque and speed area of a marine diesel engine within the limit area of the not-to-exceed zone as declared by the applicant that the engine is certified to operate within under steady -state conditions. In the case of the C1 cycle, as given by 3.2 of this Code, the not-to-exceed zone corresponds to the whole of the limit area of the not-to-exceed zone.
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1.3.30 Limit area of the not-to-exceed zone means the power or torque and speed boundaries of the not-to-exceed zone at and above 25% rated power for all test cycles as given by 3.2 of this Code except for the C1 cycle where it is at and above 50% engine load.
1.3.31 Point emission value means the NO x emission value expressed in terms

as given by 3.2 of this Code except for the C1 cycle where it is at and above 50% engine load.

1.3.31 Point emission value means the NO x emission value expressed in terms of g/kWh at the reference conditions of humidity and temperature given by this Code at a particular power or load and speed point.

1.3.32 NOx certification pack means the package of information supplied by the applicant to the Administration as required to be submitted by 2.5 and 3.3 of this Code.

1.3.33 Propulsion engine means a marine diesel engine that is used for direct or indirect propulsion. A propulsion engine may additionally perform non -propulsion duties during or separately to propulsion duties.

1.3.34 Non-propulsion engine means a marine diesel engine that is not a propulsion engine. An engine that solely or in part provides athwartships movement of a ship is not a propulsion engine.

1.3.35 Constant-speed engine means a marine diesel engine that is limited to constant-speed operation.

1.3.36 Constant-speed engine operation means a marine diesel engine regulated by a speed control device that automatically controls the operator demand to maintain

constant-speed operation.
1.3.36 Constant-speed engine operation means a marine diesel engine regulated by a speed control device that automatically controls the operator demand to maintain engine's nominal speed across the load range.*
Additionally, an idle speed setting may be provided that can be used during start -up or shutdown.

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* In service, such a speed control device may either maintain a fixed speed or a load dependent speed such that at maximum load the speed could be up to around 10% lower than at zero load.

1.3.37 Variable-speed engine means an engine that is not a constant -speed engine."
Chapter 2 Surveys and certification
3 A new section 2.5 is added as follows:
"2.5 Rational emission control strategy
2.5.1 In addition to 2.2, the requirements of this section shall apply.
2.5.2 A rational emission control strategy shall be applied to each marine diesel engine across the whole of its operating load and speed range. The means by which that is achieved shall be documented by the applicant to the Administration within a NOx certification pack. The information included in that pack shall be such as to

that is achieved shall be documented by the applicant to the Administration within a NOx certification pack. The information included in that pack shall be such as to demonstrate to the satisfaction of the Administration that a rational emission control strategy is applied during normal operation of the engine.

2.5.3 For an engine where one or more auxiliary control devices are applied, each of those shall be declared to the Administration within the NO x certification pack irrespective of whether those operate under steady -state or transient conditions. An auxiliary control device which is not so declared shall be considered a defeat device and hence invalidate the NO x certification of an engine to which such an undeclared device is applied.

2.5.4 For screening of the base emission control strategy, the NO x certification pack shall include:

.1 a list of all NO x emission influencing setting and operating values controlled by an engine's base emission control strategy, for

2.5.4 For screening of the base emission control strategy, the NO x certification pack shall include:

.1 a list of all NO x emission influencing setting and operating values controlled by an engine's base emission control strategy, for example fuel injection, inlet and exhaust valve operation, charge air management, exhaust bypass/wastegate or exhaust after-treatment controls;

.2 a record of the reference values for the settings and operating values identified in 2.5.4.1 at each of the mode points of the applicable test cycle;

.3 documentation that whenever the engine is operating between two mode points as identified in 2.5.4.2, the emission control strategy interpolates progressively between the mode points;

.4 documentation to show that, along lines of constant power and varying speed from the line between the mode points to the limit area of the not-to-exceed zone of the engine, the base emission control strategy shall ensure that any variation in the point emission values is progressive and justified from the value at that power on the line between the mode points, unless rationalized by an auxiliary

control strategy shall ensure that any variation in the point emission values is progressive and justified from the value at that power on the line between the mode points, unless rationalized by an auxiliary control device or explained by a physical limitation of the engine;
.5 a declaration that the engine's base emission control strategy only reacts to changes in engine load and speed;
.6 any other information the applicant considers relevant; and
.7 any other information the Administration requests.
2.5.5 For each auxiliary control device which may operate under steady -state conditions, the NOx certification pack shall include:
.1 a justification of the need for that device; and
.2 a description for that device, including:
.1 details of the conditions under which that device will operate and the functioning of that device;
.2 how each modulated parameter of the emission control system achieves the stated purpose of the base emission control strategy;

control strategy;

.3 the process used to ensure that the modulation is limited to the conditions where the stated purpose of the auxiliary control device operational strategy arises and to set the modulation to be the minimum necessary to achieve that stated purpose;
.4 the effect of the application of that device on the engine's base emission control strategy;
.5 for auxiliary control devices that operate above 25% engine power, the effect on the point emission values shall be documented;
.6 for auxiliary control devices that operate within the declared not-to-exceed zone, an estimate of the effect on the point emission values shall be documented;
.7 any other information the applicant considers relevant; and
.8 any other information the Administration requests.
.3 Auxiliary control devices that only operate during transient conditions need not be included in the NO x certification pack for screening.

.8 any other information the Administration requests.
.3 Auxiliary control devices that only operate during transient conditions need not be included in the NO x certification pack for screening.
2.5.6 The technical file as required by 2.3.4 shall contain the following information:
.1 identification of those auxiliary control devices declared under 2.5.3;
.2 for those auxiliary control devices covered under 2.5.5, the operating conditions which will cause those devices to function;
.3 the means by which the operation of those auxiliary control devices under 2.5.5 may be verified as part of the onboard NO x verification procedure; and
.4 where the provisions of 2.3.6 apply, the means by which it is to be verified that the required quantities of additional substance used are consistent with achieving the engine's intended base emission control strategy shall be included as part of the onbo ard NO x verification procedure.
2.5.7 Where acceptable to the Administration, the documentation requirements of 2.5.4 and 2.5.5 may alternatively be made by reference to that in respect of marine

verification procedure.
2.5.7 Where acceptable to the Administration, the documentation requirements of 2.5.4 and 2.5.5 may alternatively be made by reference to that in respect of marine diesel engines comparable, in terms of NO x emissions characteristics, to the engine to be certified.
2.5.8 The provisions of this section only apply to a marine diesel engine which is installed in a ship as an identical replacement engine if the requirements of this section applied at the time the engine family or engine group to which that engine belongs was first certified."
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Chapter 3 Nitrogen oxides emission standards
3.1 Maximum allowable NOx emission limits for marine diesel engines
4 Paragraph 3.1.4 is replaced by the following:
"3.1.4 In the case of a marine diesel engine to be certified in accordance with paragraph 5.1.1 of regulation 13, the specific emission at each individual mode point shall not exceed the applicable NOx emission limit value by more than 50% except as follows:

paragraph 5.1.1 of regulation 13, the specific emission at each individual mode point shall not exceed the applicable NOx emission limit value by more than 50% except as follows:

.1 The 10% mode point in the D2 test cycle specified in 3.2.4.

.2 The 10% mode point in the C1 test cycle specified in 3.2.5.

.3 The idle mode point in the C1 test cycle specified in 3.2.5."

3.2 Test cycles and weighting factors to be applied

5 Section 3.2 is replaced by the following:

"3.2 Test cycles and weighting factors to be applied

3.2.1 For every individual engine or parent engine of an engine family or engine group, one or more of the relevant test cycles specified in 3.2.2 to 3.2.5 shall be applied for verification of compliance with the applicable NOx emission limit contained in regulation 13. Appendix IX provides guidance on the selection of the appropriate test cycle but where discrepancies exist the text of chapter 3 takes precedence.

3.2.2 For a fixed pitch propeller propulsion engine or a propeller -law operated non-propulsion engine, test cycle E3 shall be applied in accordance with table 1.

3.2.3 For a propulsion engine that does not operate with a fixed pitch propeller,

3.2.2 For a fixed pitch propeller propulsion engine or a propeller -law operated non-propulsion engine, test cycle E3 shall be applied in accordance with table 1.
3.2.3 For a propulsion engine that does not operate with a fixed pitch propeller, including an engine fitted as part of a diesel-electric installation or an engine operated with a controllable-pitch propeller, test cycle E2 shall be applied in accordance with table 2.
3.2.4 For a non-propulsion engine that is a constant -speed engine, test cycle D2 shall be applied in accordance with table 3.
3.2.5 For a non-propulsion engine that operates as a variable -speed engine, not included above, test cycle C1 shall be applied in accordance with table 4.
Table 1 Test cycle for a marine diesel engine meeting paragraph 3.2.2
Test cycle
E3
Speed 100% 91% 80% 63%
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15
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Table 2 Test cycle for a marine diesel engine meeting paragraph 3.2.3
Test cycle
E2

Table 2 Test cycle for a marine diesel engine meeting paragraph 3.2.3

Test cycle	E2			
Speed	100%	100%	100%	100%*
Power	100%	75%	50%	25%
Weighting factor	0.2	0.5	0.15	0.15

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* There are exceptional cases, including large bore engines intended for E2 applications, in which, owing to their oscillating masses and construction, engines cannot be run at low load at nominal speed without the risk of damaging essential components. In such cases, the engine manufacturer should make an application to the Administration that the test cycle as given in table 2 above may be modified for the 25% power mode with regard to the engine speed. The adjusted engine speed at 25% power, however, should be as close as possible to the rated engine speed, as recommended by the engine manufacturer and approved by the Administration. The applicable weighting factors for the test cycle should remain unchanged.

Table 3 Test cycle for a marine diesel engine meeting paragraph 3.2.4

Test cycle	D2				
Speed	100%	100%	100%	100%	100%

The applicable weighting factors for the test cycle should remain unchanged.

Table 3 Test cycle for a marine diesel engine meeting paragraph 3.2.4

Test cycle	D2				
Speed	100%	100%	100%	100%	100%
Power	100%	75%	50%	25%	10%
Weighting factor	0.05	0.25	0.3	0.3	0.1

Table 4 Test cycle for a marine diesel engine meeting paragraph 3.2.5

Test cycle	C1							
Speed	Rated			Intermediate				Idle
Torque	100%	75%	50%	10%	100%	75%	50%	0%
Weighting factor	0.15	0.15	0.15	0.1	0.1	0.1	0.1	0.15

3.2.6 The torque figures given in test cycle C1 are percentage values that represent for a given test mode the ratio of the required torque to the maximum possible torque at this given speed.

3.2.7 The intermediate speed for test cycle C1 shall be declared by the manufacturer, taking into account the following requirements:

.1 For engines that are designed to operate over a speed range on a full load torque curve, the intermediate speed shall be the declared maximum torque speed if it occurs between 60% and 75% of rated speed.

.2 If the declared maximum torque speed is less than 60% of rated speed, then the intermediate speed shall be 60% of the rated speed.

maximum torque speed if it occurs between 60% and 75% of rated speed.

.2 If the declared maximum torque speed is less than 60% of rated speed, then the intermediate speed shall be 60% of the rated speed.

.3 If the declared maximum torque speed is greater than 75% of the rated speed, then the intermediate speed shall be 75% of rated speed.

.4 For engines that are not designed to operate over a speed range on the full load torque curve at steady -state conditions, the intermediate speed will typically be between 60% and 70% of the maximum rated speed.

3.2.8 If an engine manufacturer applies for a new test cycle application on an engine already certified under a different test cycle specified in 3.2.2 to 3.2.5, then it may not be necessary for that engine to undergo the full certification process for the new application. In this case, the engine manufacturer may demonstrate compliance by recalculation, by applying the measurement results from the specific modes of the

new application. In this case, the engine manufacturer may demonstrate compliance by recalculation, by applying the measurement results from the specific modes of the first certification test to the calculation of the total weighted emissions for the new test cycle application, using the corresponding weighting factors from the new test cycle."

6 A new section 3.3 is added as follows:

"3.3 Not-to-exceed emission values within the limit area of the not -to-exceed zone

3.3.1 The boundaries, in terms of power or torque and speed, of the not-to-exceed zone at or above 25% power shall be declared to the Administration by the applicant as part of the NO x certification pack. Operation outside these not -to-exceed zone boundaries, within the limit area of the not -to-exceed zone, shall only be permitted during starting, stopping, accelerations, deceleration, load pick -up or load rejection. However, operation below 25% power and at or above 63% speed for the E3, E2, and D2 test cycles and below 50% load for the C1 test cycle shall be permitted subject to it being shown in accordance with the requirements of 2.5 that a rational emission

and D2 test cycles and below 50% load for the C1 test cycle shall be permitted subject to it being shown in accordance with the requirements of 2.5 that a rational emission control strategy continues to be applied.

3.3.2 The technical file as required by 2.3.4 shall additionally contain the following information:

.1 the power or torque and speed boundaries, as given by 3.3.1, within which the engine is certified to operate; and

.2 the onboard NOx verification procedure shall include means to verify that the engine only operates within the power or torque and speed boundaries as given by 3.3.1.

3.3.3 Additional to the emission testing under 3.2 the Administration may, at its discretion, require that up to three point emission values be determined at load points within the not -to-exceed zone in order to verify that the not -to-exceed zone requirements are complied with. The load points to be tested shall be agreed between the applicant and the Administration as part of the review of the NOx certification pack. Point emission values are to be determined in accordance with the procedures given

the applicant and the Administration as part of the review of the NOx certification pack. Point emission values are to be determined in accordance with the procedures given by chapter 5 and appendix X. To be acceptable each of those point emission values so determined shall not exceed the respective not-to-exceed emission limit value, , as determined using the procedure in appendix X.

Point emission value emission limit value, , at that point

3.3.4 Alternative means by which it is to be shown that a point emission value may be determined or the not -to-exceed zone requirements are complied with may be used subject to their acceptability to the Administration.

3.3.5 For member engines of engine families or engine groups first certified prior to the effective date of the requirements under this section, demonstration of compliance with the requirements of this section may be on the basis solely of documentation which is to be acceptable to the Administration."

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Chapter 4 Approval of serially manufactured engines: Engine family and engine group concepts

7 In paragraph 4.3.8.2, sub -paragraphs 4.3.8.2.12 to 4.3.8.2.14 are added, after the existing sub-paragraph 4.3.8.2.11, as follows:

".12 multiple engine operational profiles as covered by chapter 8.

.13 base emission control strategy.

.14 auxiliary control devices."

8 Paragraph 4.3.10.5 is deleted.

Chapter 6 Procedures for demonstrating compliance with NOx emissions on board

9 In paragraph 6.2.2.3, at the end of sub-paragraph 6.2.2.3.15, the word "or" is deleted, at the end of sub -paragraph 6.2.2.3.16, "." is replaced with ",", and new sub-paragraphs 6.2.2.3.17 to 6.2.2.3.19 are added after sub-paragraph 6.2.2.3.16 as follows:

".17 list of identification references of all engine operational profiles available for the engine and, if applicable, the conditions under which each is to be used (see chapter 8 of the Code);

".17 list of identification references of all engine operational profiles available for the engine and, if applicable, the conditions under which each is to be used (see chapter 8 of the Code);
.18 list of auxiliary control devices accepted for the engine and the operating conditions under which those devices function; or
.19 the engine power or engine load and speed boundaries above 25% engine power within which the engine is certified to operate."

10 A new chapter 8 is added as follows:

"Chapter 8 Multiple engine operational profiles

8.1 Acceptance of multiple engine operational profiles

8.1.1 The switching between engine operational profiles under onboard conditions is permitted, subject to the provisions of this chapter, in the following cases:
.1 for a marine diesel engine certified to be in -service switchable between emission tiers;
.2 for a marine diesel engine certified to more than one test cycle application in accordance with 3.2 where the engine operational profile is in -service switchable based on the duty the engine is performing; or
.3 for a marine diesel engine certified to the same emission standard,

application in accordance with 3.2 where the engine operational profile is in -service switchable based on the duty the engine is performing; or

.3 for a marine diesel engine certified to the same emission standard, the same rated power, same rated speed and the same test cycle which is in-service switchable between multiple engine operational profiles.

8.1.2 A marine diesel engine certified in accordance with 8.1.1.1 and/or 8.1.1.2 may additionally be switchable, at a particular tier and or duty, between multiple engine operational profiles in which cases the provisions of 8.1.1.3 also apply.

8.1.3 Each engine operating profile shall be identified in the technical file as required by 2.3.4 together with the conditions, if applicable, under which each engine operating profile is to be used.

8.2 Certification of multiple engine operational profiles

8.2.1 For a marine diesel engine to which 8.1.1.1 applies, the parent engine test

operating profile is to be used.

8.2 Certification of multiple engine operational profiles

8.2.1 For a marine diesel engine to which 8.1.1.1 applies, the parent engine test report for each tier shall be included in the technical file as required by 2.4.1.5. The parent engine specific emission value for each tier shall be entered under 1.9.6 of the Supplement to the EIAPP Certificate.

8.2.2 For a marine diesel engine to which 8.1.1.2 applies, the parent engine test report for each test cycle shall be included in the technical file as required by 2.4.1.5. The test cycles for which the engine is certified shall be shown on the EIAPP Certificate. The parent engine specific emission value for each test cycle shall be entered and identified under 1.9.6 of the Supplement to the EIAPP Certificate.

8.2.3 For a marine diesel engine to which 8.1.1.3 applies:

.1 the engine test report for each engine with the parent engine features and characteristics identified in 4.3.9 or 4.4.8, for each engine operational profile, shall be determined in accordance with the provisions of chapter 5 of this Code. Where there is a mode point

features and characteristics identified in 4.3.9 or 4.4.8, for each engine operational profile, shall be determined in accordance with the provisions of chapter 5 of this Code. Where there is a mode point condition which is the same among the different engine operational profiles, that is not required to be repeated for each test cycle. The required testing may not necessarily be undertaken on the same physical engine;

.2 the specific emission value determined in accordance with 5.12.6.1 for each engine operational profile shall not exceed the applicable limit value as given by regulation 13;

.3 the multiple engine operational profile parent engine specific emission value shall be determined in accordance with 5.12.6.1 from the highest NO x emission rate, qmgasi as per 5.12.5.2, at each mode point across all the engine operational profiles for which the engine is to be certified;

.4 the parent engine test report for each engine operational profile for which the engine is to be certified shall be included in the technical file as required by 2.4.1.5 together with the determination of the multiple engine operational profile parent engi ne specific

which the engine is to be certified shall be included in the technical file as required by 2.4.1.5 together with the determination of the multiple engine operational profile parent engi ne specific emission value;

.5 the multiple engine operational profile parent engine specific emission value shall be entered under 1.9.6 of the Supplement to the EIAPP Certificate; and

.6 Section 2.2.1 of the Supplement to the IAPP Certificate shall be completed to identify which engines installed on a ship are approved to operate with multiple engine operational profiles.

8.3 Use of multiple engine operational profiles

8.3.1 An engine operational profile shall only be used in accordance with the associated conditions as given in the technical file.

8.3.2 The identification reference of the engine operational profile in use shall be recorded as part of the onboard NO x verification procedure together with data that

associated conditions as given in the technical file.
8.3.2 The identification reference of the engine operational profile in use shall be recorded as part of the onboard NO x verification procedure together with data that demonstrates that the conditions attached to the use of that engine operational profile were being complied with.
8.3.3 On change from one engine operational profile to another, the date and time of the completion of that change shall be recorded as part of the onboard NO x verification procedure for that engine."
Appendix V Parent engine test report and test data
11 The title of appendix V is replaced by the following:
"Parent engine test report, test data, and determination of the highest composite specific emission value"
12 The title of section 1 is replaced by the following:
"Section 1 Parent engine test report
(see 5.10 and 8.2 of the Code)"
13 The title of section 2 is replaced by the following:
"Section 2 Parent engine test data to be included in the technical file, additionally, for marine diesel engines to which 8.1.1.3 applies, the relevant test data for all engine operational profiles for which the engine is certified which

"Section 2 Parent engine test data to be included in the technical file, additionally, for marine diesel engines to which 8.1.1.3 applies, the relevant test data for all engine operational profiles for which the engine is certified which are to be included in the technical file (see 2.4.1.5 and 8.2 of the Code)"

14 A new section 3 is added after the existing section 2 as follows:

"Section 3 Multiple engine operational profile parent engine, determination of the composite specific emission value to be included in the technical file for engines with those multiple engine operational profiles (see 8.2 of the Code)

Calculation of the highest composite specific emission value in accordance with 8.2.3.3.

Appendix VII Checklist for an engine parameter check method

15 In paragraph 1, at the end of sub -paragraph 1.14.1, "." is replaced with ";", and sub-paragraphs 1.15 to 1.17 are added after the existing sub-paragraph 1.14 as follows:

"1.15 list of identification references of all engine operational profiles available for the engine and associated conditions, if applicable, under which each is to be used (see chapter 8 of the Code);

"1.15 list of identification references of all engine operational profiles available for the engine and associated conditions, if applicable, under which each is to be used (see chapter 8 of the Code);
1.16 list of auxiliary control devices accepted for the engine and the operating conditions under which those devices function;

1.17 the engine power or engine load and speed boundaries within which the engine is certified to operate."
16 A new appendix IX is added as follows:
"Appendix IX Flow chart for engine certification test cycle determination
(refer to 3.2 of the Code)
"
17 A new appendix X is added as follows:
"Appendix X Calculation of not-to-exceed emission limit value within not-to-exceed zone
(refer to chapters 3 and 5 of the Code)
1 This appendix describes the method for determining the not-to-exceed emission limit value, , at any point within the not-to-exceed zone for comparison with a determined point emission value as set out in 3.3 of this Code.

1 This appendix describes the method for determining the not-to-exceed emission limit value, , at any point within the not-to-exceed zone for comparison with a determined point emission value as set out in 3.3 of this Code.
2 Where engine test results are used to determine a point emission value, formula (1) shall be used to generate that value. At that point, the tolerance requirements of 5.9.6.2 of this Code apply:
 =

(1)
where:
 = NOx at the point Mn in g/kWh
 = Power at the point Mn (brake plus auxiliary) in kW
 = Mass flow rate of NOx in g/h see 5.12.5.2 of this Code
  is to be corrected for humidity and temperature consistent with the method used for the engine test from 5.12.4 of this Code.
Test cycle selection flowchart
Engine is a propulsion engine*
Engine operated with fixed pitch propeller
Yes
Yes
No
Yes
No
No
No
E2
E3
Engine is propeller-law operated
Engine is constant speed
E3
Yes D2
C1
* A propulsion engine may additionally perform non-propulsion duties during or separately to propulsion duties. An engine that solely or in part provides athwartships movement of a ship is not a propulsion engine.

E3
Yes D2
C1
* A propulsion engine may additionally perform non-propulsion duties during or separately to propulsion duties. An engine that solely or in part provides athwartships movement of a ship is not a propulsion engine.

3 Designation of not-to-exceed zone for E2, E3 and D2 test cycles in limit area of the not-to-exceed zone

3.1 The limit area of the not-to-exceed zone for engines certified to the E2 and E3 test cycles is defined by a speed boundary of equal to or greater than 63% and a power boundary of equal to or greater than 25%. The limit area of the not-to-exceed zone for the D2 cycle is defined by a power boundary of equal to or greater than 25%, at the nominal speed of the engine.

3.2 For the E3 and variable-speed application of the E2 cycle certified engines, the applicant is to define, in accordance with 3.3.1 of this Code, the not-to-exceed zone within the limit area of the not-to-exceed zone as wide or as narrow as applicable

the applicant is to define, in accordance with 3.3.1 of this Code, the not-to-exceed zone within the limit area of the not-to-exceed zone as wide or as narrow as applicable for their intended applications of the engine. The applicant -defined not-to-exceed zone shall encompass all normal steady -state speed load combinations within the limit area of the not-to-exceed zone for the applications of the engine.

3.3 The applicant's designated not-to-exceed zone can be defined by any mathematical formula(e), lists of coordinates or other method of defining the boundary. The not-to-exceed zone does not need to extend to the boundary of the limit area of the not-to-exceed zone.

3.4 For D2 and constant -speed E2 cycle certified engines, the not-to-exceed zone shall be a line of power greater than 25% at the nominal speed.

4 Determination of not-to-exceed emission limit value for E2, E3 and D2 test cycles

4.1 The not-to-exceed emission limit value at each NO x checkpoint shall be determined in accordance with the requirements of this section.

Note: If there is an auxiliary control device that causes a NOx discontinuity within the

test cycles
4.1 The not-to-exceed emission limit value at each NO x checkpoint shall be determined in accordance with the requirements of this section.
Note: If there is an auxiliary control device that causes a NOx discontinuity within the not-to-exceed zone, follow the additional procedure in section 6 to insert proxy NO x emission points to address the area(s) of discontinuity.
4.2 Interpolated NOx value at power between mode points as determined using formula (2):
= + ( )
( )
( ) (2)
where:
= Interpolated NOx value at power
= Measured point emission value according to formula (1) at nearest measured mode point at power below checkpoint power
= Measured point emission value according to formula (1) at nearest measured mode point at power above checkpoint power
= Power at checkpoint
= Power at mode point below checkpoint
= Power at mode point above checkpoint

= Power at mode point above checkpoint
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4.3 Determine the not-to-exceed emission limit value at power between the
mode points, to the Tier, as applicable.
.1 For Tier II
The not-to-exceed emission limit value at power is given by
formula (3)
= 1.2 (3)
where:
= Not-to-exceed emission limit value at power
= Interpolated NO x value at power
.2 For Tier III
The not-to-exceed emission limit value, , at power shall be
either set by 3.1.4 of this Code or as determined in accordance with
formula (4), whichever is the lower.
is the lower of or
with:
= + 0.25 (4)
= 1.5 (5)
where:
= NOx cycle limit for engine
= the maximum mode point value for the engine according
to 3.1.4 of this Code
4.4 If the checkpoint power is on the propeller law curve for an E3 certified
engine or the nominal speed line for a constant-speed E2 or a D2 certified engine:
= (6)

to 3.1.4 of this Code
4.4 If the checkpoint power is on the propeller law curve for an E3 certified engine or the nominal speed line for a constant-speed E2 or a D2 certified engine:
 = (6)
For this situation the determination of the not-to-exceed emission limit value, , is complete for that checkpoint. Otherwise, continue with 4.5.
4.5 For E3 and variable -speed application of the E2 cycle certified engines, where the checkpoint power is located at a speed not on the line between the measured mode points, carry out the additional procedure in 4.5.1 to 4.5.4.
.1 Determine the NO x limit at either edge of the not-to-exceed zone, , for the selected checkpoint power along a line of constant power, in accordance with formula (7):
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 = 1.5 (7)
with:
 =

(8)
where:
 = NOx limit at edge of not-to-exceed zone
 = NOx cycle limit for engine
 = NO x specific emission value for the engine from 5.12.6.1 of this Code

= 1.5 (7)
with:
=

(8)
where:
= NOx limit at edge of not-to-exceed zone
= NOx cycle limit for engine
= NO x specific emission value for the engine from 5.12.6.1 of this Code
.2 Determine the not-to-exceed emission limit value at a checkpoint power which is on the constant power line between the mode point line and the edge of the not-to-exceed zone in accordance with formula (9):
= + ( ) ()() (9)
with:
For tier II, = from formula (3)
For tier III, is from formula (4)
where:
= NOx limit at required checkpoint
= NOx limit at edge of not-to-exceed zone
= Speed at required checkpoint
= Speed at edge of applicants selected not-to-exceed zone at checkpoint power (may be on lower or higher side of mode line as required for value of)
= Speed on measured mode line at selected power
For engines certified to the E2 test cycle, the speed on the measured mode line, , is the nominal speed.
For engines certified to the E3 test cycle, the speed on the measured mode line, , is determined by the cube law propeller curve:
= .

3
(10)

line, , is the nominal speed.
For engines certified to the E3 test cycle, the speed on the measured mode line, , is determined by the cube law propeller curve:
 = .

3
(10)
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where:
 = Rated speed from 1.3.12 of this Code
 = Power at checkpoint
 = Rated power from 1.3.11 of this Code
.3 Determine the not-to-exceed emission limit value at power as applicable:
.1 For Tier II
The NOx limit is the interpolated result:
 = (11)
.2 For Tier III
The not-to-exceed emission limit value at power shall be set by 3.1.4 of this Code or as determined in accordance with 4.5.3.1 whichever is lower:
 is the lower of or
5 Determination of not-to-exceed emission limit value for the C1 test cycle
5.1 For the C1 test cycle within the limit area of the not-to-exceed zone, screening is conducted between the measured mode points of 100%, 75%, and 50% load at both intermediate speed (mode points 5, 6 and 7 respectively) and rated speed

cycle

5.1 For the C1 test cycle within the limit area of the not-to-exceed zone, screening is conducted between the measured mode points of 100%, 75%, and 50% load at both intermediate speed (mode points 5, 6 and 7 respectively) and rated speed (mode points 1, 2 and 3, respectively).

This creates two zones, Zone A and Zone B, where double linear interpolation or extrapolation is carried out between the nearest mode points:

.1 Zone A uses mode points 5, 1, 6 and 2. Zone A may extend above the torque line from mode point 5 and mode point 1 or beyond the speed line from mode point 1 to mode point 5.

.2 Zone B uses mode points 6, 2, 7 and 3. Zone B may extend beyond the speed line from mode point 2 to mode point 3.

.3 The applicant may request that the Administration exclude operating points from the limit area of the not-to-exceed zone screening if the applicant can demonstrate that the engine is not capable of operating at steady state at those points when installed on a ship. Otherwise, the not-to-exceed zone consists of the entire limit area of the not-to-exceed zone.

5.2 Determine if the checkpoint is in Zone A or Zone B by determining if the

on a ship. Otherwise, the not-to-exceed zone consists of the entire
limit area of the not-to-exceed zone.
5.2 Determine if the checkpoint is in Zone A or Zone B by determining if the
checkpoint torque, , is higher or lower than the torque on the boundary between
Zone A and Zone B (75% load line) for the checkpoint speed.
 = 6 + ( ) (62)() (12)
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where:
 = Torque at checkpoint speed on a straight line between mode point 6
and mode point 2 (75% load line)
6 = Torque at mode point 6 (75% of torque at intermediate speed)
2 = Torque at mode point 2 (75% of torque at rated speed)
 = Checkpoint speed
 = Intermediate speed
 = Rated speed
5.3 Determine the interpolated/extrapolated NOx value at desired checkpoint:
.1 If the checkpoint torque, , is greater than , use equation (13) and
(14) for the interpolation or extrapolation.
 = + ( ) ()() (13)
with:
 = 5 + ( ) (51)() (14)
where:

.1 If the checkpoint torque, , is greater than , use equation (13) and (14) for the interpolation or extrapolation.

 = + ( ) ()() (13)

with:

 = 5 + ( ) (51)() (14)

where:

1 = Torque at mode point 1 (100% of torque at rated speed)

5 = Torque at mode point 5 (100% of torque at intermediate speed)

 = Torque at checkpoint speed on a straight line between mode point 6 and mode point 2 (75% load line) from formula (12)

 = Torque at checkpoint speed on a straight line between mode point 5 and mode point 1 (100% load line) from formula (14)

 = Torque at checkpoint

 = Interpolated NOx at checkpoint speed on 100% load line

 = Interpolated NOx at checkpoint speed on 75% load line

 = Checkpoint speed

 = Intermediate speed

 = Rated speed

.2 If the checkpoint torque, , is less than use equation (15) and (16) for the interpolation or extrapolation.

 = + ( ) ()() (15)

with:

= + ( ) ()() (15)

with:

= 7 + ( ) (73)() (16)

where:

3 =Torque at mode point 3 (50% of torque at rated speed)

7 =Torque at mode point 7 (50% of torque at intermediate speed)

=Torque at checkpoint speed on a straight line between mode point 6 and mode point 2 (75% load line) from formula (12)

=Torque at checkpoint speed on a straight line between mode point 7 and mode point 3 (50% load line) from formula (16)

=Torque at checkpoint

=Interpolated NOx at checkpoint speed on 75% load line

=Interpolated NOx at checkpoint speed on 50% load line

= Checkpoint speed

=Intermediate speed

=Rated speed

5.4 Determine the not-to-exceed emission limit value at checkpoint:

.1 For Tier II

The not-to-exceed emission limit value is given by formula (17):

= 1.2 (17)

where:

= Not-to-exceed emission limit value at checkpoint

= Interpolated NOx value at power

.2 For Tier III

.1 For Tier II
The not-to-exceed emission limit value is given by formula (17):
 = 1.2 (17)
where:
 = Not-to-exceed emission limit value at checkpoint
 = Interpolated NOx value at power
.2 For Tier III
The not-to-exceed emission limit value shall be either set by 3.1.4 of this Code or as determined in accordance with 5.3, whichever is the lower:
 is the lower of or
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with:
 = + 0.25 (18)
 = 1.5 (19)
where:
 = Interpolated NOx value at checkpoint
 = NOx cycle limit for engine
 = the maximum mode point value according to 3.1.4 of this Code
6 Method to address discontinuity in the operation zone due to an auxiliary control device
6.1 For each approved auxiliary control device, where there is operation in the not-to-exceed zone that causes a discontinuity in the NO x emissions it can be necessary to introduce additional proxy mode points to account for the discontinuity

6.1 For each approved auxiliary control device, where there is operation in the not-to-exceed zone that causes a discontinuity in the NO x emissions it can be necessary to introduce additional proxy mode points to account for the discontinuity in the area of engine operation where that auxiliary control device is active.
6.2 There will be two or more proxy mode points to cover the action of an auxiliary control device.
6.3 is calculated in the same manner as 4.2 using the proxy points where necessary in the interpolation.
6.4 Use good engineering judgement that may include simulation or in -house testing to determine the appropriate NOx level and location of the proxy points.
6.5 The engine power of the proxy mode points may overlap to account for hysteresis that may occur as a result of approaching the points from rising or falling power. The overlap should also take into account any variation in the operating point of the auxiliary control device based on engine speed.
6.6 Include the proxy mode points as part of the auxiliary control device documentation supplied to the Administration in the NOx certification pack."

of the auxiliary control device based on engine speed.
6.6 Include the proxy mode points as part of the auxiliary control device documentation supplied to the Administration in the NOx certification pack."
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